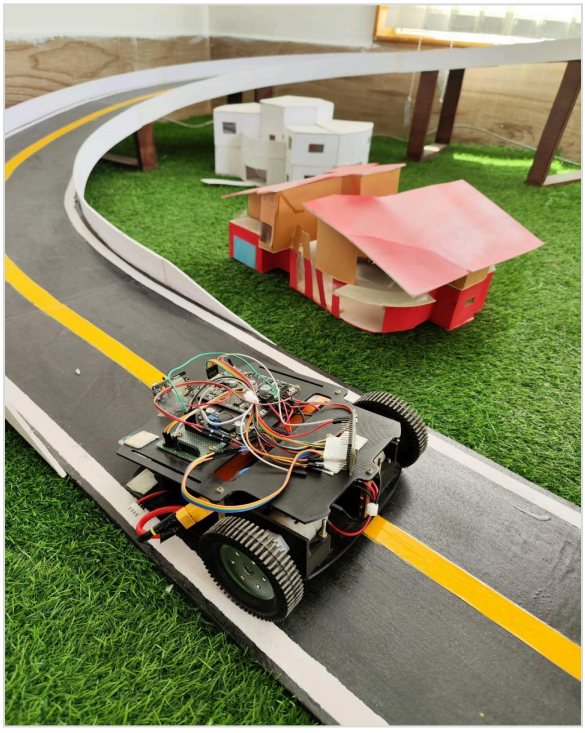
IEEE RAS PESU 2023

Robofest 3.0 took place from October 18th to November 19th, 2023, at the MRD Auditorium, PES University. The event began with an inauguration on October 18th in front of the BE Block, **"Unveiling of the banner with a drone reveal and a performance by Team Trance"**.The contest saw a total of 75 teams registering, with 23 teams of 1-4 members actively participating. The project showcase occurred on October 26th and 27th in front of the BE Block.

Following this, the **Online National ReRo Lab Hackathon** was held on November 3rd and 4th, 2023 via Discord and the ReRo Hackathon website, attracting 27 registered teams who competed in IoT and ROS-based rounds. The hackathon presented problem statements focused on developing innovative IoT solutions and creating efficient algorithms using the Robot Operating System (ROS).

Problem statements included tasks such as

- controlling an autonomous UGV to navigate an arena autonomously using ROS2 and Gazebo simulation software (ARTPARK),
- designing a computer vision-enabled robotic arm for retail store shelves (PVL),
- modeling a robotic system for pick and place tasks (cRAIS),
- and an open innovation category encouraging participants to actively seek ideas and resources for robotics and automation (Open Innovation).





A club orientation was conducted on November 6th, accompanied by pre-hackathon workshops focused on ROS and CAD Modelling, led by student mentors.

The event culminated in a **24-hour offline hackathon** on November 18th and 19th, 2023 at the MRD Auditorium, PES University. This intensive session included multiple tracks: ARTPARK, PVL, cRAIS, and Open Innovation, each with unique problem statements challenging participants to solve real-world problems through robotics and AI solutions. Judging began at 8 AM on November 19th, followed by final evaluations and the elimination of three teams.

ARTPARK @IISc and PESU Venture Labs (PVL) sponsored the event.

The winners:

- **Team "Bananas"** for the ARTPARK Track
- **Bhupinder Jogi's Team** for the PVL Track
- **Team "Chaotic Bots"** for the cRAIS Track
- **Team "Guru AI"** for the Open Innovation Track